Stochastic Finance and Risk Modelling

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1 - From probability to stochastic process

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Outline

Probability measure

Independence

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$\sigma-$algebra

Definition (measurable space)

Let Ω be a set and $\mathcal{F}$ be a collection of subsets of Ω, i.e. $\mathcal{F} \subseteq 2^{\Omega} := \{A : A \subset \Omega\}$. Then $\mathcal{F}$ is said to be a $\sigma-$algebra on Ω if

- $\emptyset \in \mathcal{F}$;
- $A \in \mathcal{F} \implies A^c := \Omega \setminus A \in \mathcal{F}$;
- $(A_n)_{n\geq 1} \subset \mathcal{F} \implies \cup_{n\geq 1} A_n \in \mathcal{F}$.

$(\Omega, \mathcal{F})$ is called a measurable space.

- Examples :
  1. Trivial sigma-algebra $\mathcal{F} = \{\emptyset, \Omega\}$.
  2. Sigma-algebra of all events : the power set $\mathcal{F} = 2^{\Omega}$.
  3. Sigma-algebra associated to a partition $(A_i)_{i\in I}$ i.e. $\Omega = \bigsqcup_{i\in I} A_i$. Then $\mathcal{F} = \{\bigsqcup_{j\in J} A_j : J \in \mathcal{P}(I)\}$ is the smallest sigma-algebra that contains the $(A_i)_{i\in I}$.

Generated σ-algebra

Definition

For any $\mathcal{C} \subseteq 2^{\Omega}$, there exists a smallest sigma-algebra on Ω –in the sense of the inclusion– which contains $\mathcal{C}$. It is called the sigma-algebra generated by $\mathcal{C}$ and denoted by $\sigma(\mathcal{C})$:

$$\sigma(\mathcal{C}) := \bigcap_{\mathcal{F}:\ \sigma\text{-algebra},\ \mathcal{F}\supset\mathcal{C}} \mathcal{F}.$$

Recall that if $(\mathcal{F}_\lambda)_{\lambda\in\Lambda}$ is a family of σ-algebras, $\cap_{\lambda\in\Lambda}\mathcal{F}_\lambda$ is also a σ-algebra.

Borel σ-algebra

- On a topological space Ω, the Borel sigma-algebra $\mathcal{B}(\Omega)$ is the one generated by $\mathcal{O}$ the family of open sets of Ω.
- In $\mathbb{R}$, the Borel sigma-algebra is the smallest one that contains the intervals.
- In $\mathbb{R}^d$, it is generated by the the sets
$$\prod_{i=1}^{d}]a_i, b_i[= \{\omega = (\omega_1, \ldots, \omega_d) \in \mathbb{R}^d \ : \ \omega_i \in]a_i, b_i[, \ i = 1, \ldots, d\}.$$
- $\mathcal{B}(\mathbb{R}^d)$ is very big but there exists some subsets of $\mathbb{R}^d$ which are not Borel.

Product σ-algebra

Let $\left((\Omega_n, \mathcal{F}_n)\right)_{n\geq 1}$ be a sequence of measurable spaces.

- Finite product sigma-algebra on $\Omega_1 \times \ldots \times \Omega_n$: $\mathcal{F}_1 \otimes \ldots \otimes \mathcal{F}_n$.
- It is the smallest sigma-algebra that contains $\mathcal{F}_1 \times \ldots \times \mathcal{F}_n$.
- Note that $\mathcal{B}(\mathbb{R}) \otimes \ldots \otimes \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^n)$.
- Product sigma-algebra $\Pi_{n\geq 1}^{\otimes}\mathcal{F}_n$ on $\Pi_{n\geq 1}\Omega_n$: smallest σ-algebra generated by the collection of cylinders with finite basis $\mathcal{C}$:

$$\mathcal{C} = \{\prod_{i\geq 1} A_i \ : \ A_i \in \mathcal{F}_i \text{ and } A_i = \Omega_i \text{ except for a finite number of index}\}$$

Measurable function

Definition (Measurable function)

For measurable spaces $(\Omega, \mathcal{F}), (\Sigma, \mathcal{G})$, the map $f : \Omega \to \Sigma$ is said to be measurable if

$$f^{-1}(B) := \{\omega \in \Omega : f(\omega) \in B\} \in \mathcal{F}, \quad \forall B \in \mathcal{G}.$$

Here, $f^{-1}(B)$ denotes the inverse image of B by f.

- For $(f_\lambda)_{\lambda \in \Lambda}$ with $f_\lambda : (\Omega, \mathcal{F}) \to (\Sigma_\lambda, \mathcal{G}_\lambda)$, set $\sigma(f_\lambda : \lambda \in \Lambda) := \sigma(\mathcal{C}_\Lambda)$, where

$$\mathcal{C}_\Lambda \ := \ \{f_\lambda^{-1}(B) : \ B \in \mathcal{G}_\lambda, \lambda \in \Lambda\}.$$

- $\sigma(f_\lambda : \lambda \in \Lambda)$ is the "smallest" algebra making f_λ $\mathcal{F} - \mathcal{G}_\lambda$ measurable for every $\lambda \in \Lambda$.

Probability measure

Definition (measure)

A map $\mu : \mathcal{F} \to [0, \infty]$ is called a measure if

- $\mu(\emptyset) = 0$;
- $\mu(\sqcup_{n\geq 1} A_n) = \sum_{n\geq 1} \mu(A_n)$ for any $(A_n)_{n\geq 1} \subset \mathcal{F}$ of disjoint sets.

μ is called a probability measure if $\mu(\Omega) = 1$. We will denoted by $\mathcal{P}(\Omega, \mathcal{F})$ the set of all probability measures on $(\Omega, \mathcal{F})$. For $\mu \in \mathcal{P}(\Omega, \mathcal{F})$, $(\Omega, \mathcal{F}, \mu)$ is called a probability space.

Definition (pushforward measure)

Let $\mu \in \mathcal{P}(\Omega, \mathcal{F})$ and $f : \Omega \to \Sigma$ be measurable. The pushforward measure $f_{\#}\mu$ is defined by $f_{\#}\mu(B) := \mu(f^{-1}(B))$ for all $B \in \mathcal{G}$.

It is a measure on $(\Sigma, \mathcal{G})$. In the french notation $f_{\#}\mu = \mu_f$ the image law of μ by f.

Support

Definition (support)

The support of a measure μ, if it exists, is a closed set, denoted by $supp(\mu)$, satisfying $\mu(\Omega \setminus supp(\mu)) = 0$ and if G is open and $G \cap supp(\mu) \neq \emptyset$, then $\mu(G \cap supp(\mu)) > 0$.

If Ω is a Polish space (separable completely metrizable topological space) and μ is a finite measure defined on $\mathcal{B}(\Omega)$, then μ admits a unique support and

$$supp(\mu) = \Omega \setminus \big(\cup \{V : V \text{ open and } \mu(V) = 0\}\big).$$

We say that μ is supported in $K \in \mathcal{F}$ if $\mu(\Omega \setminus K) = 0$. So, if Ω is a Polish space and μ is a finite measure defined on $\mathcal{B}(\Omega)$, $supp(\mu)$ is the smallest closed subset in which μ is supported.

Random variable

Definition (random variable)

Let $\Sigma = \mathbb{R}^d$ and $\mathcal{G} = \mathcal{B}(\mathbb{R}^d)$. Any measurable function $X : \Omega \to \mathbb{R}^d$ is called a r.v. on $(\Omega, \mathcal{F})$.

For every $\mathbb{P} \in \mathcal{P}(\Omega, \mathcal{F})$, $\mathrm{P}_X := X_\# \mathbb{P} \in \mathcal{P}(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ denotes the law/distribution of X under $\mathbb{P}$. For all $A \in \mathcal{B}(\mathbb{R}^d)$

$$\mathrm{P}_X[A] = \mathbb{P}[X \in A].$$

Definition (expectation)

Fix $\mathbb{P} \in \mathcal{P}(\Omega, \mathcal{F})$. A r.v. $X : \Omega \to \mathbb{R}^d$ is integrable if $\int_\Omega |X(\omega)| \mathbb{P}(d\omega) < \infty$.

Denote the expectation of X by

$$\mathbb{E}[X] := \int_\Omega X(\omega) \mathbb{P}(d\omega).$$

Related terminology

Definition ($L^p_d(\Omega, \mathcal{F}, \mathbb{P})$)

For each $p > 0$, denote by $L^p_d(\Omega, \mathcal{F}, \mathbb{P})$ the space of r.v.s $X : \Omega \to \mathbb{R}^d$ s.t. $|X|^p$ is integrable.

When $d = 1$, we just write $L^p(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (variance/covariance)

Let $d = 1$. If $X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, denote by $Cov(X, Y)$ the covariance between X and Y

$$Cov(X, Y) := \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Define further the variance of X by $Var(X) := Cov(X, X)$.

Theorem of transport

Theorem

For measurable spaces $(\Omega, \mathcal{F})$ *and* $(\Sigma, \mathcal{G})$*, fix* $\mathbb{P} \in \mathcal{P}(\Omega, \mathcal{F})$*. Let* $X : \Omega \to \Sigma$ *be* $\mathcal{F} - \mathcal{G}$ *measurable and* $g : \Sigma \to \mathbb{R}^d$ *be* $\mathcal{G} - \mathcal{B}(\mathbb{R}^d)$ *measurable. Denote* $Y = g \circ X : \Omega \to \mathbb{R}^d$*. Then* $Y \in L^1_d(\Omega, \mathcal{F}, \mathbb{P})$ *iff* $g \in L^1(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mathrm{P}_X)$*. In this case,*

$$\mathbb{E}[g(X)] = \int_\Omega g \circ X d\mathbb{P} = \int_{\mathbb{R}^d} g(x) \mathrm{P}_X(dx).$$

List of common distributions

Let $supp(X) := supp(\mathrm{P}_X)$.

1. X (resp. P_X) has (resp. is) binomial distribution $\mathcal{B}(n,p)$ if $supp(X) = \{0,\dots,n\}$ and
$$\mathbb{P}[X=k] = \mathrm{P}_X(\{k\}) = \frac{n!}{k!(n-k)!}p^k(1-p)^{n-k}, \quad k=0,\dots,n;$$
2. X has Poisson distribution $\mathcal{P}(\lambda)$ if $supp(X) = \mathbb{N}$ and
$$\mathrm{P}_X[\{k\}] = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k=0,1,\dots;$$
3. X has Gaussian distribution $\mathcal{N}(m,\sigma^2)$ if $supp(X) = \mathbb{R}$
$$\mathrm{P}_X(dx) = \frac{1}{\sqrt{2\pi}\sigma}e^{-(x-m)^2/2\sigma^2}dx;$$
4. X has uniform distribution $\mathcal{U}(a,b)$ if $supp(X) = [a,b]$ and
$$\mathrm{P}_X(dx) = \frac{\mathbf{1}_{[a,b]}(x)}{b-a}dx.$$

Outline

Probability measure

Independence

Convergence

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Equivalent probability measures

Definition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and Λ some non empty set. The events $(A_\lambda)_{\lambda \in \Lambda}$ of $\mathcal{F}$ are independent (mutually independent) if for all $k \geq 1$, $\lambda_1 < \ldots < \lambda_k$ in Λ thus FOR ALL FINITE SUB-FAMILY

$$\mathbb{P}(A_{\lambda_1} \cap \ldots \cap A_{\lambda_k}) = \mathbb{P}(A_{\lambda_1}) \ldots \mathbb{P}(A_{\lambda_k}).$$

- Events can be independent two by two (pairwise independent) but not independent.
- Let $\Omega = \{\omega_1, \ldots, \omega_4\}$, $\mathcal{F} = 2^\Omega$:
- $\mathbb{P}(\{\omega_i\}) = 1/4$ for $i \in \{1, \ldots, 4\}$.
- Let $A_1 = \{\omega_1, \omega_4\}$, $A_2 = \{\omega_2, \omega_4\}$ and $A_3 = \{\omega_3, \omega_4\}$.
- A_1 and A_2 are independent, A_2 and A_3 are independent, A_1 and A_3 are independent since

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1 \cap A_3) = \mathbb{P}(A_2 \cap A_3) = \mathbb{P}(\{\omega_4\}) = \frac{1}{4}$$

$$\mathbb{P}(A_i)\mathbb{P}(A_j) = \left(\frac{1}{4} + \frac{1}{4}\right)\left(\frac{1}{4} + \frac{1}{4}\right) = \frac{1}{4}.$$

- BUT A_1, A_2, A_3 are not independent since

$$\mathbb{P}(A_1 \cap A_2 \cap A_3) = \frac{1}{4} \text{ and } \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3) = \frac{1}{8}.$$

Independence

Fix $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (independence of algebras)

The σ-algebras $\mathcal{G}_1, \ldots, \mathcal{G}_n \subset \mathcal{F}$ are independent if

$$\mathbb{P}(\cap_{i=1}^n A_i) = \Pi_{i=1}^n \mathbb{P}(A_i), \quad \forall A_i \in \mathcal{G}_i,\ 1 \leq i \leq n.$$

A family of σ-algebras $(\mathcal{G}_\lambda)_{\lambda \in \Lambda}$ is said to be independent if $G_{\lambda_1}, \ldots, G_{\lambda_n}$ are independent for all $\lambda_1, \ldots, \lambda_n \in \Lambda$ and $n \geq 1$.

Definition (independence of r.v.s)

A family of r.v.s $(X_\lambda)_{\lambda \in \Lambda}$ with $X_\lambda : (\Omega, \mathcal{F}) \to (\Sigma_\lambda, \mathcal{G}_\lambda)$ is independent if $(\sigma(X_\lambda))_{\lambda \in \Lambda}$ is independent i.e. for all $n \geq 1$ and $\lambda_1, \ldots, \lambda_n \in \Lambda$., for all $A_i \in \Sigma_{\lambda_i}$, $i = 1 \ldots n$,

$$\mathbb{P}\left(\{X_{\lambda_1} \in A_1\} \cap \ldots \cap \{X_{\lambda_n} \in A_n\}\right) = \mathbb{P}(\{X_{\lambda_1} \in A_1\}) \ldots \mathbb{P}(\{X_{\lambda_n} \in A_n\}).$$

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Convergence

Definition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $(Y_n)_{n\in\mathbb{N}^*}$ and Y be some $\mathbb{R}^d$-valued random variables. Then $(Y_n)_{n\in\mathbb{N}^*}$ converges to Y,

- $\mathbb{P}$ almost surely ($\mathbb{P}$-a.s.) $Y_n \xrightarrow[n\to+\infty]{\text{a.s.}} Y$ if

$$\mathbb{P}\left[\left\{\omega \in \Omega \ : \ |Y_n(\omega) - Y(\omega)| \underset{n\to\infty}{\longrightarrow} 0\right\}\right] = 1.$$

- in $L^p_d(\Omega, \mathcal{F}, \mathbb{P})$ $(1 \leq p < +\infty)$ $Y_n \xrightarrow[n\to+\infty]{L^p} Y$ if for all $n \in \mathbb{N}$ $Y_n, Y \in L^p_d(\Omega, \mathcal{F}, \mathbb{P})$ and

$$\|Y_n - Y\|_p \ := \ (\mathbb{E}\left[|Y_n - Y|^p\right])^{1/p} \underset{n\to\infty}{\longrightarrow} 0.$$

Convergence

Definition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $(Y_n)_{n\in\mathbb{N}^*}$ and Y be some $\mathbb{R}^d$-valued random variables. Then $(Y_n)_{n\in\mathbb{N}^*}$ converges to Y,

- in probability $Y_n \xrightarrow[n\to+\infty]{\text{proba.}} Y$ if for $\epsilon > 0$, $\mathbb{P}\left[|Y_n - Y| > \epsilon\right] \xrightarrow[n\to\infty]{} 0$.
- in law $Y_n \xrightarrow[n\to+\infty]{\text{law}} Y$ if and only if for all bounded and continous function $f : \mathbb{R}^d \to \mathbb{R}$, $\mathbb{E}(f(Y_n)) \xrightarrow[n\to+\infty]{} \mathbb{E}(f(Y))$.

For the last convergence only the law of the random variables matter. For example, if $Y \sim \mathcal{N}(0,1)$ then one can replace it by any random variable with the same law and write $Y_n \xrightarrow[n\to+\infty]{\text{law}} \mathcal{N}(0,1)$.

More theorems

Let $(X_n)_{n\geq 1}$ be a sequence of independant identically distributed **i.i.d. r.v.s on** $(\Omega, \mathcal{F}, \mathbb{P})$**.**

Theorem (Law of large numbers)

Assume $X_n \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ *and set* $m := \mathbb{E}[X_1]$*. Then*

$$\bar{X}_n := \frac{X_1 + \cdots + X_n}{n} \xrightarrow[n\to+\infty]{a.s.} m$$

Theorem (Central limit theorem)

Assume further $X_n \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ *and set* $\sigma^2 := Var(X_1)$*. Then*

$$\frac{\sum_{k=1}^n (X_k - m)}{\sqrt{n}\sigma} = \sqrt{n}\frac{\bar{X}_n - m}{\sigma} \xrightarrow[n\to+\infty]{law} \mathcal{N}(0,1).$$

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Conditional expectation in L^1

Theorem (Existence and uniqueness)

Let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G} \subset \mathcal{F}$. There is a unique $\mathcal{G}-$measurable r.v., denoted by $\mathbb{E}[X|\mathcal{G}] \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and called the conditional expectation of X knowing $\mathcal{G}$, s.t.

$$\mathbb{E}[XZ] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}]Z], \quad \forall \textit{bounded and } \mathcal{G}\textit{-mesurable } Z.$$

We may replace Z by $\mathbf{1}_A$ for $A \in \mathcal{G}$. Further, if $\mathcal{G} = \sigma(Y_\lambda : \lambda \in \Lambda)$, we write $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X|Y_\lambda, \lambda \in \Lambda]$.

Properties of conditional expectation

Proposition

For $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$*, one has :*

(a) *(tower law property)* $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[X|\mathcal{H}]$ *a.s. for all* σ*-algebras* $\mathcal{H} \subset \mathcal{G} \subset \mathcal{F}$ *;*

(b) *(linearity)* $X \mapsto \mathbb{E}[X|\mathcal{G}]$ *is a linear map ;*

(c) *(positivity) If* $X \geq 0$ *a.s., then* $\mathbb{E}[X|\mathcal{G}] \geq 0$ *a.s. ;*

(d) *(representation)* $\mathbb{E}[X|Y] = g(Y)$ *a.s. for some measurable function* g*.*

Properties of conditional expectation

Proposition

Let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ *and* $\mathcal{G} \subset \mathcal{F}$.

(a) $\mathbb{E}[X|\mathcal{G}] = X$ *a.s. iff* X *is* $\mathcal{G}$*-measurable.*

(b) *If* X *is independent of* $\mathcal{G}$*, then* $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$ *a.s.*

(c) *Let* Y *is* $\mathcal{G}$*-measurable s.t.* $XY \in L^1(\Omega, \mathcal{F}, \mathbb{P})$*. Then* $\mathbb{E}[XY|\mathcal{G}] = Y\mathbb{E}[X|\mathcal{G}]$ *a.s.*

(d) *If* $\varphi : \mathbb{R} \to \mathbb{R}$ *is convex, then* $\varphi\left(\mathbb{E}[X|\mathcal{G}]\right) \leq \mathbb{E}[\varphi(X)|\mathcal{G}]$ *a.s.*

Theorem

Let $(X_n)_{n\geq 1} \subset L^1(\Omega, \mathcal{F}, \mathbb{P})$ *converge almost surely to* X*. Then* $\lim_{n\to\infty} \mathbb{E}[X_n|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}]$ *almost surely if one of the following conditions holds*

(i) $|X_n| \leq Z \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ *for all* $n \geq 1$*;*

(ii) $X_n \geq 0$ *increases to* X *almost surely.*

A special case : $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$

Note that $L^2(\Omega, \mathcal{F}, \mathbb{P})$ is a Hilbert space endowed with the scalar product $\langle X, Y \rangle := \mathbb{E}[XY]$.

Proposition

Let $\mathcal{G} \subset \mathcal{F}$.

(a) *$L^2(\Omega, \mathcal{G}, \mathbb{P})$ is a closed subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$.*

(b) *$\mathbb{E}[X|\mathcal{G}]$ is identified as the projection of X in $L^2(\Omega, \mathcal{G}, \mathbb{P})$, i.e.*

$$\langle X - \mathbb{E}[X|\mathcal{G}], Z \rangle = 0, \quad \forall Z \in L^2(\Omega, \mathcal{G}, \mathbb{P}).$$

Gaussian vector

Definition

A random vector $X = (X_1, \cdots, X_d)$ of $\mathbb{R}^d$ is said to be gaussian if

- any linear combination of its components is a gaussian real valued random variable.
- $\forall a \in \mathbb{R}^d$, $a^t X = \sum_{i=1}^{d} a_i X_i$ is a gaussian real valued random variable.
- A gaussian law μ on $\mathbb{R}^d$ is a law of a gaussian vector i.e. $\mu = \mathbb{P}_{(X_1,\ldots,X_d)}$.

The components X_i of a gaussian vector X are always gaussian real valued random variables but the converse is false in general.

Regression

Theorem

Let $(Y_1, \ldots, Y_n, X)$ *be a gaussian vector of* $L^2(\Omega, \mathcal{F}, \mathbb{P})$*. Then there exist* $\Lambda = (\lambda_1, \ldots, \lambda_n)^t \in \mathbb{R}$ *such that*

$$\mathbb{E}[X|Y_1, \ldots, Y_n] = \mathbb{E}[X] + \sum_{j=1}^{n} \lambda_j (Y_j - \mathbb{E}(Y_j)) = \mathbb{E}[X] + \Lambda^t (Y - \mathbb{E}[Y]),$$

where

$$\Lambda = \Sigma_Y^{-1} \begin{pmatrix} cov(X, Y_1) \\ \vdots \\ cov(X, Y_n) \end{pmatrix}.$$

Moreover $X - \mathbb{E}[X|Y_1, \ldots, Y_n]$ *is independent of* $\sigma(Y_1, \ldots, Y_n)$*.*

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Change of measure/Bayes formula

Let $Z \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ be positive a.s. and $\mathbb{E}[Z] = 1$. Define $\mathbb{Q} \in \mathcal{P}(\Omega, \mathcal{F})$ as follows : For any $A \in \mathcal{F}$, set

$$\mathbb{Q}(A) := \mathbb{E}_{\mathbb{P}}[1_A Z] = \int_A Z(\omega) d\mathbb{P}(\omega).$$

- Then $\mathbb{Q} \in \mathcal{P}(\Omega, \mathcal{F})$.
- $\mathbb{Q}$ is a probability measure with density Z with respect to $\mathbb{P}$.
- Z is called the derivative of Radon-Nikodym of $\mathbb{Q}$ with respect to $\mathbb{P}$, denoted by $Z = \dfrac{d\mathbb{Q}}{d\mathbb{P}}$ or $\mathbb{Q} = Z.\mathbb{P}$.
- For any real valued random variable (or $\mathbb{R}^d$ random vector) X on Ω $X \in L^1_d(\Omega, \mathcal{F}, \mathbb{Q})$ if and only if $XZ \in L^1_d(\Omega, \mathcal{F}, \mathbb{P})$. In this case

$$\mathbb{E}_{\mathbb{Q}}[X] = \mathbb{E}_{\mathbb{P}}[XZ]$$

  Works also if $d = 1$ and $X \geq 0$ $\mathbb{P}$-a.s.

Conditional Bayes formula

Theorem

- *Let* $(\Omega, \mathcal{F}, \mathbb{P})$ *be a probability space,* $\mathcal{G}$ *be a sub-sigma-algebra of* $\mathcal{F}$ *and* $Z : \Omega \to \mathbb{R}_+^*$ *a random variable such that* $\mathbb{E}_{\mathbb{P}}[Z] = 1$.
- *Let* $\mathbb{Q}$ *be defined by* $\mathbb{Q}[A] = \mathbb{E}_{\mathbb{P}}[1_A Z] = \int_\Omega 1_A(\omega) Z(\omega) \mathbb{P}(d\omega)$ *for all* $A \in \mathcal{F}$.
- $\mathbb{Q}$ *is a probability measure with density* Z *with respect to* $\mathbb{P}$.
- *Notation* $Z = \dfrac{d\mathbb{Q}}{d\mathbb{P}}$ *or* $\mathbb{Q} = Z.\mathbb{P}$.
- *For any real valued random variable (or* $\mathbb{R}^d$ *random vector)* X *on* Ω, $X \in L_d^1(\Omega, \mathcal{F}, \mathbb{Q})$ *if and only if* $XZ \in L_d^1(\Omega, \mathcal{F}, \mathbb{P})$. *In this case*

$$\mathbb{E}_{\mathbb{Q}}[X \mid \mathcal{G}] = \frac{\mathbb{E}_{\mathbb{P}}[XZ \mid \mathcal{G}]}{\mathbb{E}_{\mathbb{P}}[Z \mid \mathcal{G}]} \quad \mathbb{P}\text{-a.s.}$$

Works also if $d = 1$ *and* $X \geq 0$ $\mathbb{P}$*-a.s.*

Equivalent probability measures

Let $\mathbb{P}, \mathbb{Q}$ be two probability measures on $(\Omega, \mathcal{F})$.

Definition (Absolutely continuous probability measures)

$\mathbb{Q}$ is absolument continue w.r.t. $\mathbb{P}$, denoted by $\mathbb{Q} \ll \mathbb{P}$ if and only if for all $A \in \mathcal{F}$ $\mathbb{P}(A) = 0 \Longrightarrow \mathbb{Q}(A) = 0$.

Definition

$\mathbb{P}$ and $\mathbb{Q}$ are equivalent if $\mathbb{Q} \ll \mathbb{P}$ and $\mathbb{P} \ll \mathbb{Q}$. This is denoted by $\mathbb{P} \sim \mathbb{Q}$. The two measures have the same impossible (resp. sure) events.

- Let $\mathbb{Q}$ be a probability measure with density Z with respect to $\mathbb{P}$ then $\mathbb{Q} \ll \mathbb{P}$. Indeed let $A \in \mathcal{F}$ such that $\mathbb{P}[A] = 0$ then $\mathbb{Q}[A] = \int 1_A Z d\mathbb{P} = 0$.
- Assume further $\mathbb{P}[Z > 0] = 1$, then $\mathbb{P}$ and $\mathbb{Q}$ are equivalent.

Radon-Nykodym Theorem

Theorem

Let $\mathbb{Q}$ and $\mathbb{P}$ be two probability measures on $(\Omega, \mathcal{F})$. Then the following assertions are equivalent

1. $\mathbb{Q} \ll \mathbb{P}$.
2. *There exists $f \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ $f \geq 0$ (unique a.s.) such that $\mathbb{Q}$ has a density function f with respect to $\mathbb{P}$, i.e. $\mathbb{Q} = f \cdot \mathbb{P}$ or* $\mathbb{Q}[A] = \int 1_A f d\mathbb{P} = \mathbb{E}[1_A f]$ *or* $\dfrac{d\mathbb{Q}}{d\mathbb{P}} = f$.